

# **International Semiconductor Research and Development Capabilities: An Analysis Framework and Interactive Database**

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## **Executive Summary**

This report, conducted as part of the Department of State's Virtual Student Foreign Service program, describes the development of and key takeaways from an interactive map of global semiconductor research and development, with a particular focus on countries outside the United States and China, which many other sources have already covered. The map highlights key technology focus areas, research capabilities, and innovation clusters around the world. In addition to the map itself, the report also establishes a reusable research and analysis methodology for studying the semiconductor industry.

The report provides background information on the semiconductor value chain, research capabilities, and key technology areas. It then dives into a detailed global and regional analysis, presenting a snapshot of the 624 institutions represented in the database, including breakdowns by institution type, value chain position, research areas, and facility capabilities. Regional trends are examined, highlighting the dominance of EUR and EAP in high-performing semiconductor research institutions. The report also identifies 19 potential innovation clusters across the globe based on their concentration of universities, industry, and other institutions with strong collaborative ties and patent output.

The methodology section provides an in-depth look at the data sources, processing techniques, and visualization tools used to create the map and analysis. This includes Institute of Electrical and Electronics Engineers (IEEE) publications and U.S. Patent Office (USPTO) patents as primary data sources, as well as automated data processing and manual classification to analyze institutions, research focus areas, and collaborations. The data from these sources was collected in January 2024.

The report concludes by discussing potential future improvements to the methodology, such as expanded data processing capabilities, increased geographical scope, and greater coverage of governmental and commercial entities. Overall, the report aims to be an important tool for understanding the global semiconductor research and development landscape.

# Introduction

This report has two aims: first, to map semiconductor research and development around the globe, highlighting technology focus areas and research capabilities; and second, to create a research and analysis methodology for future adaptation and use. In support of these goals, this project focuses primarily outside the two highest-visibility countries (the United States and China) broadening the scope to impactful research elsewhere, highlighting the global nature of semiconductor cooperation, and avoiding repetition of work carried out by other sources.

The report uses many classification criteria to detail an institution's semiconductor research, but three specifically will be highlighted here: value chain focus areas, research capabilities, and research areas. A value chain focus area is a classification that groups institutions based on their positions in the production process. The four value chain buckets are:

- design including (materials design, architectural design, and security and IP),
- manufacturing and assembly including (advanced packaging and advanced design),
- applications including (AI and computing, energy, manufacturing, and defense), and
- basic research including (advanced materials and heterogeneous applications).

Research capabilities is a high-level assessment of an institution's on-site laboratory equipment and research capacity, broken into five categories: modeling and simulation and design, fabrication, wafer and die measurement and metrology, end-use integration and test, and other capabilities. Lastly, research areas describe an institution's technological focus among a list of high-priority technologies: applications, emerging technologies, design, architecture, storage and memory, fabrication and manufacturing, and AI and machine learning.<sup>1</sup> The information in this report was collected beginning in January 2024.

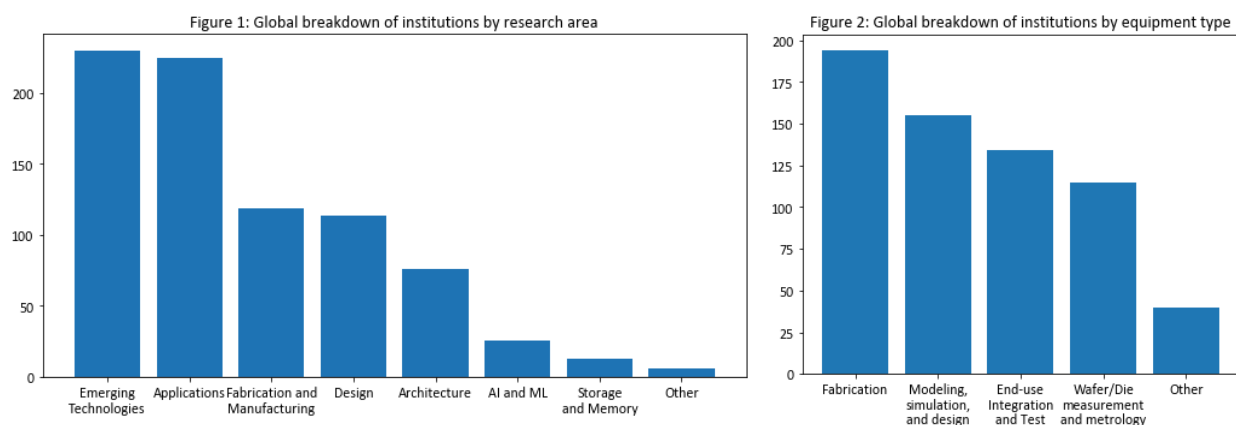
## Global and Regional Analysis

### Global Snapshot

This database represents 624 distinct institutions. By institution type, these include 360 universities, 180 companies (30 of which are satellites), 54 non-profits, and 30 government affiliated. Universities are overrepresented due to the academic focus within the data sources used, namely the IEEE publications. Institutions occupy different roles across the value chain, and in some cases hold more than one position, including 427 in basic research, 244 in applications, 211 in design, and 101 in manufacturing.

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<sup>1</sup> These categories reflect the methodology employed by the NSF in its 2022 RFI for information on semiconductor research and education. For more details, see the methodology section of this report, and the original NSF report, Sankar Basu and Erik Brunvand, *Summary of Responses to NSF/CISE Request for Information on Semiconductor Research and Education*, Alexandria: National Science Foundation, 2022.



Institutional research facilities were labeled by types of equipment services, and 365 research facilities represented multiple services, including 199 in fabrication, 154 in modeling, simulation, and design, 136 in end-use integration and test, 116 in wafer and die measurement and metrology, and 40 in other.

## Regional Trends

The majority of high-performing institutions are located in EUR and EAP, with those two regions accounting for 84.6% of total institutions. In absolute terms, EAP accounts for the majority of institutions focused on design, manufacturing, and applications, while EUR accounts for the majority of institutions focused on basic research. Regions also tend to specialize in specific components of the value chain, with EUR and AF on basic research, WHA on applications, and SCA and NEA on design. EAP institutions are active across all four components of the value chain. In terms of research areas, EUR has the most institutions participating in every category of research except Fabrication; this includes AI/ML (77%), applications (71%), architecture (61%), design (57%), emerging technologies (52%), and storage and memory (69%), reflecting the region's strength in basic research at the university level across diverse topics. EAP has the next most institutions for the research areas of AI/ML, applications, storage and memory, and fabrication, while WHA is second in architecture and design.

|                       | AF | EAP | EUR | NEA | SCA | WHA |
|-----------------------|----|-----|-----|-----|-----|-----|
| AI/ML                 | 0  | 4   | 20  | 0   | 1   | 1   |
| Applications          | 2  | 27  | 159 | 3   | 13  | 21  |
| Architecture          | 0  | 7   | 46  | 3   | 9   | 11  |
| Design                | 1  | 8   | 64  | 3   | 18  | 20  |
| Emerging technologies | 1  | 100 | 120 | 0   | 2   | 7   |
| Fabrication           | 1  | 58  | 46  | 2   | 2   | 10  |
| Storage and memory    | 0  | 2   | 9   | 0   | 1   | 1   |

Table 1: Regional breakdown of research areas

Lastly, in terms of research facilities, EUR leads across end-use integration and test, fabrication, and metrology, while EAP leads in modeling, simulation, and design. NEA, SCA, and WHA institutions have diverse facilities, with a focus on modeling, simulation, and design.

|                                     | AF | EAP | EUR | NEA | SCA | WHA |
|-------------------------------------|----|-----|-----|-----|-----|-----|
| End-use integration and testing     | 2  | 16  | 95  | 2   | 7   | 13  |
| Fabrication                         | 1  | 28  | 145 | 1   | 9   | 15  |
| Modeling, simulation, and design    | 0  | 60  | 54  | 5   | 17  | 18  |
| Wafer/die measurement and metrology | 0  | 1   | 93  | 1   | 10  | 11  |

Table 2: Regional breakdown of research facilities

## Clusters

Clusters of innovation represent localized geographies exhibiting high interconnectedness between universities, industry, and other stakeholders,<sup>2</sup> and offer the greatest potential for sustaining both basic and applied research. The report identifies the 19 cities below as potential clusters of innovation due to their combination of universities and industry (as well as other institutions), with a clear record of cooperation and high patent output. They are candidates for further research, looking at factors including shared laboratories, enabling regulations, government funding, strong human capital, and high degrees of research commercialization.<sup>2</sup>

| Country        | City      | # of Institutions | Country     | City      | # of Institutions |
|----------------|-----------|-------------------|-------------|-----------|-------------------|
| Japan          | Tokyo     | 27                | India       | Bangalore | 5                 |
| France         | Paris     | 19                | Germany     | Berlin    | 5                 |
| South Korea    | Seoul     | 15                | Ireland     | Dublin    | 5                 |
| Singapore      | Singapore | 14                | South Korea | Icheon    | 5                 |
| Taiwan         | Hsinchu   | 12                | Japan       | Kanagawa  | 5                 |
| France         | Grenoble  | 7                 | Taiwan      | Kaohsiung | 5                 |
| United Kingdom | London    | 7                 | Japan       | Kawasaki  | 5                 |
| Canada         | Ottawa    | 7                 | Australia   | Sydney    | 5                 |
| South Korea    | Daejeon   | 6                 | France      | Toulouse  | 5                 |
| South Korea    | Suwon     | 6                 |             |           |                   |

<sup>2</sup> *Clusters, Innovation and Entrepreneurship*. Paris: OECD, 2009.

Table 3. Global clusters by country, primary city, and number of institutions in primary city

## Methodology

This project draws on various data sources, software packages, database management tools as well as manual data collection and classification techniques. This section briefly covers the notable aspects of this process, with further technical documentation available in the technical supplemental (linked in the Appendix).

### Data Sources

Three data sources form the primary basis of the research: the Institute of Electrical and Electronics Engineers (IEEE) Xplore database, the U.S. Patent Office database (USPTO), and manual research, which are discussed at length in the Classification section.

IEEE is the leading global publisher<sup>3</sup>, by volume and impact, focused on academic electronics research. The IEEE Xplore database features broad coverage in topics, global scope, prestige, and overall comprehensiveness in terms of the academic ecosystem relating to semiconductors. For example, IEEE claims to publish around a third of all electrical engineering-related publications yearly. IEEE also receives publications from around the world and has resources to encourage non-English language authors to be published, allowing for a wider subset of papers and institutions to be considered.

Since devices must receive U.S. patents to be protected in the United States regardless of their country of origin, the USPTO's data has global representation even though the office is national in focus, and in this report, it serves as a proxy for fully international patent data. Additionally, the Patent Cooperation Treaty (PCT) allows for coverage of many international patents in the U.S. USPTO also has a free API that allows for easy querying of data.

The process for sampling data from these sources is largely the same outside of the specific technical process. Both data sources are queried for papers/patents with 'semiconductor' in the title, as well as various other associated keywords. In the case of patents, the provided descriptions were also included. This was not the case for IEEE papers, as many electrical engineering papers may include a reference to a potential application to semiconductors without actually being directly related. This process is designed to capture the broadest possible set of papers that are likely to be relevant and allows for processing later on to remove data sources that were deemed irrelevant. This criteria included a minimum 100 citation count per paper threshold, the removal of duplicates and non-article publications (i.e. conference papers) that were erroneously included in the dataset, and a removal of all papers older than 2000. These

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<sup>3</sup> *IEEE at a Glance*. IEEE, 2024.

restrictions were placed to create a focus on the most relevant and important papers that were likely to lead to the inclusion of the highest-performing institutions and to create a meaningful limit on the sheer workload required to assess such a large amount of data. Both sets of data were retrieved and processed using the programming language Python (for a complete technical accounting of the data process and specific codes used see the companion document).

Next, the raw data is automatically checked for relevance by checking either IEEE keywords for papers or International Patent Classification (IPC) codes that indicate the patent was in a semiconductor-related field. After processing and removal of exact duplicates, approximately 42000 papers and 50000 patents remained.

## Data Utilization

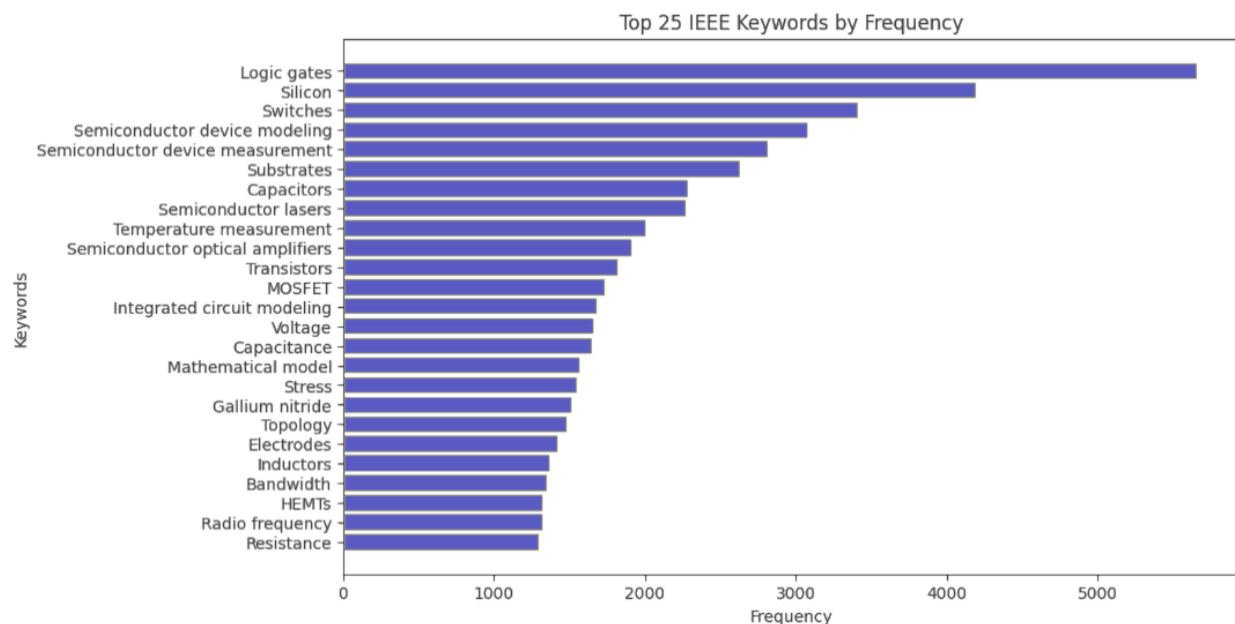


Figure 3. IEEE keywords by frequency

Figure 3 shows the distribution of IEEE keywords and provides a rough overview of the most important topics in academia, which helped inform the team's ideas about where to next focus efforts. Additionally, the aggregate citation counts from papers were grouped by both institution and country, with the latter being more successful in assigning every paper to an associated country or countries based on the affiliation of the authors. This dataset also provides data on collaborations between institutions and countries. Again, the latter grouping was more successful, and the former will be elaborated on in a later section. To focus the data on the most impactful papers and corresponding institutions and to reduce the size of the dataset to a manageable scale, the final dataset only included publications with at least 100 citations. As the intent is to focus on the leading institutions in a given country or region, selected exceptions are present in countries and regions whose largest institution was close to the threshold.

Additionally, for all countries that were present in the dataset but had no institutions represented at all, a notable institution was added based on a combination of total citation count and other qualitative factors. The result of this analysis was a country-grouped list of institutions from both IEEE and USPTO.

## Manual Collection and Classification

This report's data sources included institutional statistics such as aggregate citation counts and international collaborations automatically, whereas location, institution type, research areas, and various other data required manual collection. To standardize data that was collected manually, our dataset is organized in terms of data importance tiers, specific information classes, and a standard data collection template. Table 4 lists the fields with the corresponding importance tier and the source where the data point is typically found.

| Tier            | Data Point                                                               | Source                                                |
|-----------------|--------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Tier I</b>   | Institution Name                                                         | Provided from IEEE/USPTO data                         |
|                 | State/Province                                                           | Manual research                                       |
|                 | City                                                                     | Manual research                                       |
| <b>Tier II</b>  | Sub-institutions (colleges, etc.) (list)                                 | Provided from IEEE/USPTO data or from manual research |
|                 | Research facilities (list)                                               | Provided from IEEE/USPTO data or from manual research |
|                 | Research areas (list)                                                    | Manual research and classification                    |
|                 | Value chain                                                              | Manual research                                       |
|                 | Institution type                                                         | Manual research and classification                    |
|                 | Contact information                                                      | Manual research                                       |
| <b>Tier III</b> | Funding sources, collaboration types, amounts, and collaborations (list) | Manual research and classification                    |
|                 | R&D capabilities (e.g., design, fabrication, modeling, simulation, etc.) | Manual research and classification                    |

Table 4. Data type by source and tier classification

The various standardized lists (research capabilities, value chain, etc.) were chosen to be consistent with similar references that covered these topics. For the value chain groupings, two primary sources include a 2021 BCG and Semiconductor Industry Association report<sup>4</sup> and *The East Asian Computer Chip War*<sup>5</sup> by Monique Chu. For research areas, the primary sources were

<sup>4</sup> *Strengthening the Global Semiconductor Supply Chain in an Uncertain Era*, Boston: Boston Consulting Group and Semiconductor Industry Association, 2021

<sup>5</sup> Monique Chu, *The East Asian Computer Chip War*, New York: 2013

a 2022 NSF RFI<sup>6</sup> and a White House report on semiconductor leadership<sup>7</sup>. For R&D capabilities, both the NSF and BCG/SIA reports were consulted.

## Data Visualization

The manual data after being validated for completeness was loaded from Excel into a SQL (Structured Query Language) database for long-term data storage and ease of access. The data visualization tool Tableau was used to generate the dashboard and country-specific graphics, with global and regional graphics generated in Python.

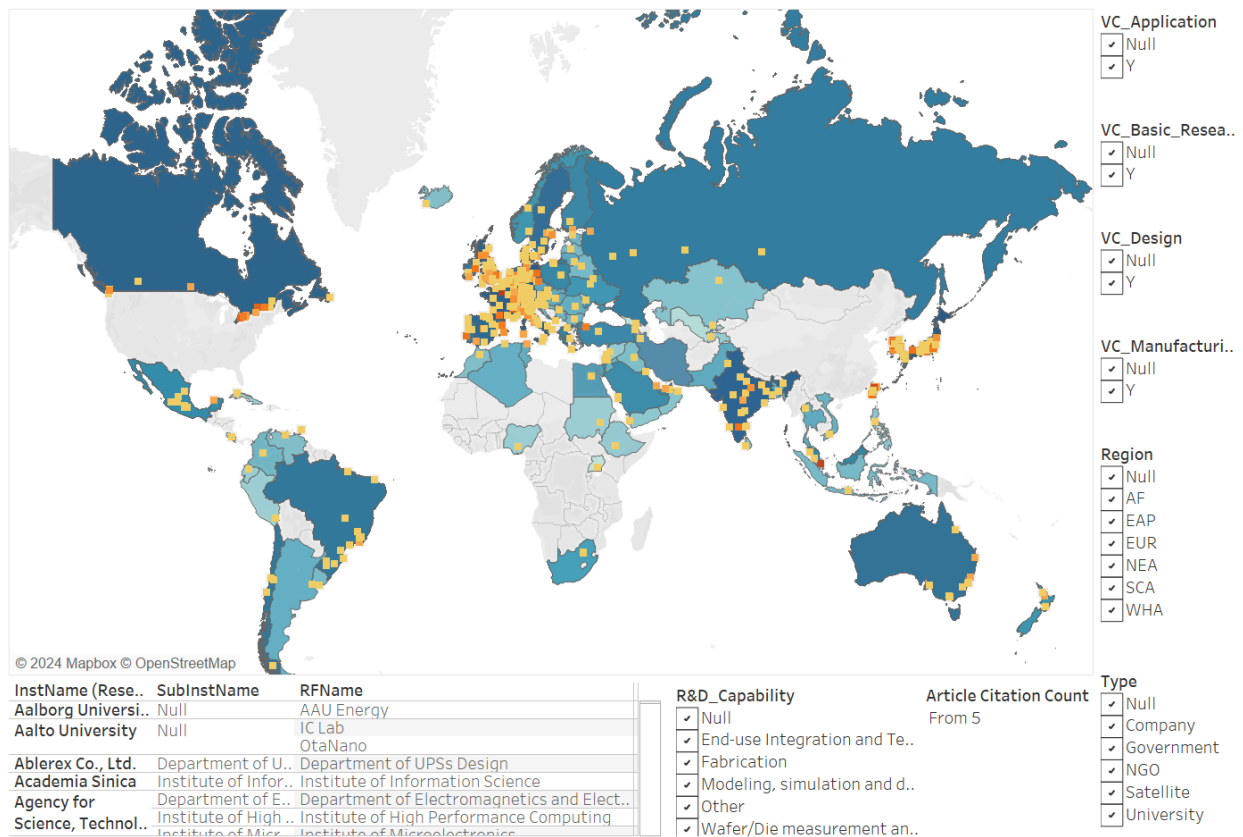


Figure 4. Image of the Tableau visualization

Figure 4 shows the visualization in Tableau. The main feature is an interactive map of the world, showing a node for each city with at least one institution, colored on a logarithmic scale corresponding to the number of institutions present. Countries are also colored on a logarithmic scale corresponding to the total number of citations from that country as measured in the IEEE data. The map also features tooltips for both countries and cities, with the former showing

<sup>6</sup> Sankar Basu and Erik Brunvand, *Summary of Responses to NSF/CISE Request for Information on Semiconductor Research and Education*, Alexandria: National Science Foundation, 2022.

<sup>7</sup> *Report to the President: Ensuring Long-Term U.S. Leadership in Semiconductors*, Washington D.C.: President's Council of Advisors on Science and Technology, 2017



aggregate statistics like total citation count, number of institutions, and various category breakdowns. The latter shows a list of each institution in the selected city, with the corresponding type. Clicking on a city also filters the chart in the bottom right to show all affiliated substitutions corresponding to the institutions in that city. Lastly, various filters are present on the right side of the dashboard. The first four filter by cities with an institution carrying out work in a certain part of the value chain, and the last filters by region.

## Future Improvements

The process laid out here exemplifies how policymakers can utilize open-source data available to survey critical industries and make data-based decisions. Despite the work carried out, there are still various improvements that a team with more resources (time, money, and more) could make.

The primary improvement that a future team could make is to invest more time and manpower in the data processing side. For example, limited time to resolve inconsistencies in institutions' names, such as abbreviations or alternative spellings, limited the team's ability to account for all the institutions in the data set and prevented aggregating statistics in terms of citation count or paper-specific metrics. Another addition could be the automatic geolocating of institutions, allowing for a much faster and more scalable product. One notable path is the use of machine learning or artificial intelligence to rapidly decrease the time needed to match data points, such as Natural Language Processing (NLP) or loading the data into a generative AI model.

Another area for improvement is geographical scope. Despite efforts to cover a broad geographical area, many developing countries and newcomers to the semiconductor space were not fully covered due to time constraints. The previous suggestion would help, but the inclusion of foreign-language or region-specific journals or patent databases would dramatically increase the specificity in these areas.

Lastly, more coverage of non-academic sources would provide a vital commercial aspect to this project that at present is not fully explored. The nature of IEEE being a more useful and comprehensive source than USPTO naturally pushed the team towards an academic focus, but a successor team could rectify this. Some ideas include foreign patent databases, startup and venture capital databases like Crunchbase, national industry associations, or foreign contacts such as embassies to request data on foreign companies from national governments.

The above suggestions are by no means exhaustive, and many more technical and data-related additions could be made to this project. However, the work that has been completed provides an initial base of open-source data about the semiconductor ecosystem as well as a technical foundation for future teams to add to.

# Conclusion

This report and accompanying database represent a tool for policymakers across the interagency to better understand global semiconductor innovation. It allows insights into thematic trends in basic and applied research, as well as geographic trends across regions, countries, and cluster cities. It therefore aims to support efforts to invest in advanced semiconductor R&D and facilitate partnerships between the U.S., partner countries, and advanced research institutions.

With the existing database and methodology included in this report, this database represents an evergreen product, capable of being regularly updated and refined. As mentioned in the methodology section, future upgrades could include additional data processing to include broader geographic regions and non-academic sources. It could also be upgraded to detail countries and institutions benefiting under existing partnerships and funding from the U.S. government, as well as other details to support USG goals. The potential clusters identified could benefit from further investigation, including a more detailed methodology to assess their potential and inform targeted investments and funds.

The semiconductor innovation ecosystem, much like the industry itself, is global and segmented. The findings from this report and the accompanying database can equip policymakers to map key institutions, research focuses, and R&D capabilities, in support of efforts to advance collaborative, leading-edge innovation in semiconductors.

# Appendix

## One-Page Summaries

To provide an example application, the team has used the interactive database to produce one-page summaries of key semiconductor R&D statistics for Japan, Taiwan, and South Korea in EAP; the Netherlands, the United Kingdom, and Switzerland in EUR; Brazil and Canada in WHA; and India in SCA.

## Technical Supplement

The companion document *International Semiconductor Research and Development Capabilities: Technical Details Supplement* provides in-depth information about the technical tools and processes used to generate the results here. This document serves to facilitate the reproducibility of results, encourage data-sharing practices, and enable the work carried out here to be extended by future teams.